ViT-GAN

- ViT
- GAN
  - vanilla GAN → Progressive GAN
  - WGAN → Improved WGAN → Spectral Normalization
  - f-GAN
- ViTGAN
  - Discriminator → ViT + WGAN + Spectral Norm
  - Gererator ☆ → z as condition
    - patch embedding to pixel.
      - ☆ siren
        - implicit neural representation

Prior

$z \sim N(0, I)$ → $G(z)$

$D = \{x_1, x_2, \dots, x_n\}$

$\tilde{D} = \{G(z_1), G(z_2), \dots, G(z_n)\}$

divergence

$KL(p \| q) = \int_x p(x) \log \frac{p(x)}{q(x)} dx.$

p_G, q P_{data}

$JS(p \| q) = \frac{1}{2} KL(p \| M) + \frac{1}{2} KL(q \| M)$

$\forall x,\ M(x) = [p(x) + q(x)]/2.$

$JS(P_G \| P_{data}) = \frac{1}{2} KL(P_G \| \frac{P_G + P_{data}}{2}) + \frac{1}{2} KL(P_{data} \| \frac{P_G + P_{data}}{2})$

$= \frac{1}{2} \int_x \left[P_G(x) \log \frac{P_G(x)}{\frac{P_G(x) + P_{data}(x)}{2}} + P_{data}(x) \log \frac{P_{data}(x)}{\frac{P_G(x) + P_{data}(x)}{2}} \right] dx$

$= \frac{1}{2} \int_x \left[P_G(x) \log \frac{P_G(x)}{P_G(x) + P_{data}(x)} + P_{data}(x) \log \frac{P_{data}(x)}{P_G(x) + P_{data}(x)} \right] dx + \log 2$

x → $D(x)$ → 0/1

$0 \to P_G$

$1 \to P_{data}$

$D^*(x) = \frac{P_{data}(x)}{P_{data}(x) + P_G(x)}$ ✓☆

① train $D(x)$ with MLE to get

$D^*(\cdot) = \underset{D(\cdot)}{\operatorname{argmax}} \left\{ E_{x \sim P_{data}}[\log D(x)] + E_{x \sim P_G}[\log(1 - D(x))] \right\}$

$\int_x P_{data}(x) \log D(x) + P_G(x) \log(1 - D(x))\, dx$

② train $G(z)$ with

$G^*(\cdot) = \underset{G(\cdot)}{\operatorname{argmin}} \left\{ E_{x \sim P_{data}}[\log D^*(x)] + E_{x \sim P_G}[\log(1 - D^*(x))] \right\}$

$f(y) = a \log y + b \log(1 - y)$

$\frac{a}{a+b}$

$\int_x P_{data}(x) \log \frac{P_{data}(x)}{P_{data}(x) + P_G(x)} + P_G(x) \log \frac{P_G(x)}{P_{data}(x) + P_G(x)}\, dx$

$\underset{G(\cdot)}{\operatorname{argmin}} \left\{ \underset{f(\cdot) \in 1\text{-lipschitz}}{\operatorname{argmax}} \left\{ E_{x \sim P_{data}}[f(x)] - E_{x \sim P_G}[f(x)] \right\} \right\}$

$F(x)$ $\quad \forall x_1, x_2 \quad |F(x_1) - F(x_2)| \le K \times |x_1 - x_2|$

① ViT ✓ (1 2 3 4)

② ViT × (0 1 2 3 4)

③ z → ✓✓ (1 2 3 4)

NN → grid → base

标题	年
Implicit Neural Representations with Periodic Activation Fun…	2020
Generative Adversarial Networks	2014
Improved Training of Wasserstein GANs	2017
Wasserstein GAN	2017
PROGRESSIVE GROWING OF GANS FOR IMPROVED QUALIT…	2018
Which Training Methods for GANs do actually Converge?	2018
Spectral Normalization for Generative Adversarial Networks	2018
ViTGAN: Training GANs with Vision Transformers	2021
TransGAN: Two Transformers Can Make One Strong GAN	2021
Combining Transformer Generators with Convolutional …	2021
Generative Adversarial Transformers	2021